

the project business team. Feedback can be structured or unstructured. The structured feedback can be seen as application reviews over various platforms or surveys. Unstructured feedback is seen through users' feedback over various social media channels. Gathering the feedback connects users with the team. It helps them enhance and address the user experience in subsequent releases.

Continuous monitoring

Continuous monitoring is crucial for assessing compliance issues and security threats in the application. It involves continuous monitoring of storage, networks, or server notifications. This early analysis helps the DevOps team detect and divert threats and take appropriate measures to reduce the downtime of the application. The entire process is automated in the DevOps pipeline, and monitoring is done using various tools to ensure high reliability and stability of the complete setup.

Continuous operations

After the successful configuration of the production environment and deployment of the application, the main challenge is to keep the application running without any hiccups 'Continuous operations' focuses on maintaining a smoothly running application — there should be minimal to no downtime. Maintenance is a major part of continuous operations, ensuring that the recent releases, patches, and updates do not impact the current deployment in any way.

The team motive is delivered through this phase because it automatically handles the network, database, and application configurations needed in the production environment. This leads to synchronised deployment and constant feedback. It enables scaling up and down based on requirements as well as rollover to the previous state if something undesirable happens.

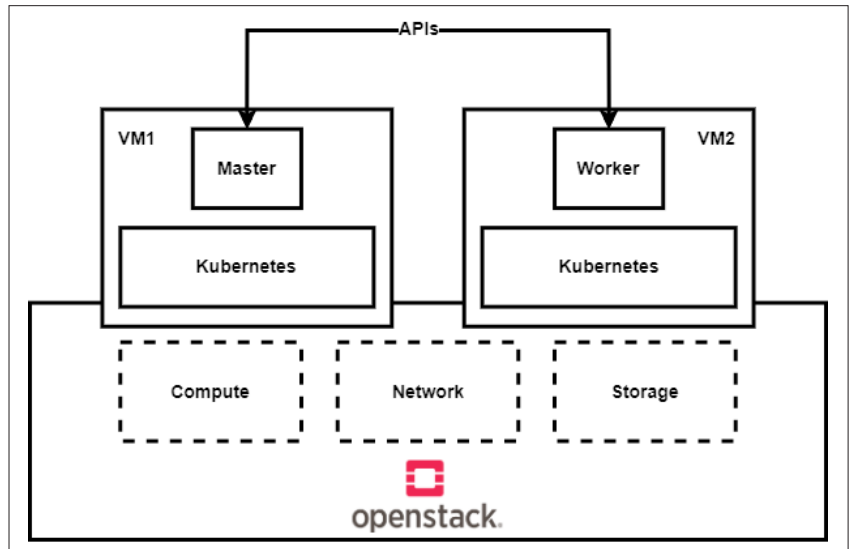


Figure 4: Kubernetes cluster hosted by OpenStack

Kubernetes enables continuous operations. It offers features like rollback, rollout, self-healing, and node monitoring, effectively delivering a robust environment with minimum downtime.

Additionally, OpenStack is used for infrastructure deployment, providing a platform for Kubernetes as shown in Figure 4. OpenStack serves the purpose of IaaS where the organisation's data is confidentially stored in the in-house cloud environment enabling flexibility over the computation needs.

Adopting DevOps enables organisations to deliver their products

at a faster pace. It promotes automation and the cultural changes that foster collaboration among multidisciplinary teams, ensuring accountability for the product. DevOps equips teams with in-house capabilities to address challenges and bail out teams in case of issues. The integration, deployments, and operations of the products can be easily done without any help from external teams. By following the phases of DevOps outlined above, organisations can achieve rapid and quality deployments with higher security and minimal or zero downtime. **END** 🐧

References

- Len Bass, Ingo Weber, and Liming Zhu, DevOps: A Software Architect's Perspective, Addison-Wesley Professional, 1st edition, 2015. ISBN: 978-0-13-404984-7
- Alanna Brown, N Forsgren, Jez Humble, Nigel Kersten, and Gene Kim, State of DevOps report, Puppet+ DORA, 2016
- Christof Ebert, Gorka Gallardo, Josune Hernantes, and Nicolas Serrano, DevOps. IEEE Software, 33(3):94–100, 2016. DOI:10.1109/MS.2016.68
- Hiren Dhaduk, DevOps lifecycle: 7 phases explained in detail with examples, <https://www.simform.com/blog/devops-lifecycle/>. Last accessed: Mar. 19, 2022
- GitHub, <https://github.com/>. Last accessed: Mar. 19, 2022
- Jenkins, <https://www.jenkins.io/>. Last accessed: Mar. 19, 2022
- Red Hat Ansible automation platform, <https://www.ansible.com/>. Last accessed: Mar. 19, 2022

By: Shubham Aggarwal, Nithya Ganesan, and B. Thangaraju

The authors are associated with the Open Source Technology Lab at the International Institute of Information Technology, Bengaluru.